

A physics-informed machine learning framework for unified prediction of superconducting transition temperatures

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ABSTRACT

Superconducting materials hold transformative potential for quantum technologies and energy systems. Still, their discovery is hindered by the complexity of traditional computational methods and the limited scope of existing machine learning (ML) models. We introduce a physics-informed (PI) machine learning framework that predicts critical transition temperatures (T_c) across all superconductor families, including both conventional and unconventional ones. By engineering 33 quantum-mechanically PI descriptors (e.g., CuO_2 layers for cuprates, Fe layer separation for iron-based systems, and electron concentration for conventional superconductors and ...), besides the compositional descriptors, our “SuperConNet” classifier achieves near-perfect 99.94 % accuracy in categorizing superconductors and precise T_c prediction. Trained on 17,820 known superconductors, which we labeled in four classes and validated against unseen materials from the Inorganic Crystal Structure Database (ICSD), our approach outperforms BCS theory and density functional theory (DFT) in both speed and generality. Crucially, SHAP analysis reveals how descriptors isolate pairing mechanisms, while DFT cross-validation confirms physical soundness. This work spans materials chemistry with quantum physics, modifying the search for high- T_c superconductors beyond narrow material classes.

1. Introduction

Superconductivity has emerged as a central topic in materials science and physics [1]. Due to their unique properties, superconducting materials play a crucial role in advanced technologies [2,3], including electronic components [4], quantum computing [5–11], and sensing devices [12–14]. However, a significant challenge in this field is identifying materials with high critical transition temperatures (T_c). Currently, research primarily focuses on discovering superconductivity in metal hydride compounds under extremely high pressures (on the order of several hundred gigapascals),^[15–22] which limits the range of materials under investigation [23,24].

Nevertheless, the discovery of superconducting materials with low transition temperatures remains significant [25–27]. Beyond their applications in near-ground-state quantum processes at millikelvin temperatures, such as superconducting qubits [11,28–30], these materials could provide new insights into the mechanisms of Cooper pair formation [31–34]. Understanding and engineering these mechanisms could,

in turn, lead to the discovery of new superconductors. However, experimentally exploring a wide range of compounds is challenging [35, 36], particularly for those with complex crystal structures and complicated fabrication processes.

BCS theory is one of the most widely used approaches for investigating superconductivity in materials [37,38]. This theory explains that electron-phonon interactions [39] lead to the formation of Cooper pairs, causing the material to expel all magnetic fields from its interior when cooled below its critical temperature—a phenomenon known as the Meissner effect—thereby enabling superconductivity [27]. Additionally, BCS equations provide tools for calculating the transition temperature of conventional superconductors. Based on this framework, superconductors are generally classified into two categories: conventional superconductors [40], which adhere to BCS theory and its extensions, and unconventional superconductors [41], which exhibit superconductivity through alternative mechanisms [42].

However, for conventional superconductors, the extensive computational demands of BCS theory limit its applicability to large datasets

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[43,44]. Moreover, this method is restricted to predicting conventional superconductors [45], whereas many materials with high transition temperatures do not follow the electron-phonon coupling mechanism [46–48]. In some cases, BCS theory even yields results that contradict experimental data, highlighting its limitations for certain compounds.

In recent years, machine learning and artificial neural networks have emerged as powerful tools for predicting new materials and their specific properties [49–57], and complementing or replacing traditional methods such as BCS and complex density functional theory (DFT) calculations [43,58–62]. These approaches enable the exploration of a broader range of materials and can predict superconducting properties with significantly greater speed and reasonable accuracy [63–67]. Studies have demonstrated that deep learning models can compute superconducting properties much faster than traditional methods [68–70]. Furthermore, integrating neural networks with DFT has been investigated as an effective approach for discovering new superconducting materials. Additionally, machine learning and neural networks have been utilized to analyze the superconducting properties of cuprates [71,72].

Despite these advancements, current research still faces notable limitations [73]. Many machine learning models are confined to binary classification or specific classes of materials with general features in all classes, such as cuprates and/or Fe-based [72,74–77], organic superconductors [78], or other narrowly defined categories. This highlights the need to develop more comprehensive and generalizable models that can analyze a broader range of materials, including both conventional and unconventional superconductors.

In this study, we constructed a general model by generating physics-informed descriptors and physical features to predict the transition temperature in materials. First, we classified the synthesized inorganic superconducting materials and extracted their transition temperatures into four groups (cuprates, simple Fe-based, complex Fe-based, and other superconducting structures) for analysis. Our results demonstrate that the method can predict the class and transition temperature of superconductors, both in conventional and unconventional compounds. The algorithm embeds three fundamental superconducting mechanisms: a) charge transfer physics and subsequent spin fluctuation dynamics (cuprates), implemented via O/Cu ratio and Hole doping based on Zhang-Rice singlet model [79] (hole concentration $\propto [O]/([Cu] + [O])$). b) Spin-fluctuation mediation (iron-based), characterized by magnetic

exchange coupling (J) and Fe-Fe distance (d), and c) Conventional electron-phonon coupling, quantified by the average electron-phonon coupling constant (λ) and the average phonon frequency (ω), plays a fundamental role in BCS theory and its extensions. This method represents the crucial first component encoding fundamental superconductivity physics into computable features that enable efficient machine learning for materials discovery. The PI-ML workflow is shown in Fig. 1. This Figure shows our workflow in detail.

2. Data preparation

This study uses two main datasets to build and test our machine learning model. First, we took data from the SuperCon database [80], which started with about 21,200 superconducting materials. We carefully cleaned this data by: 1) removing entries with unclear chemical formulas (like generic “Cuprate” labels), 2) excluding high-pressure hydrides (hydrogen-rich with high $T_c > 30$ K), and 3) filtering out poorly defined entries. This left us with 17,820 reliable superconductors with confirmed critical temperatures (T_c), covering four material classes. Second, we used the Inorganic Crystal Structure Database (ICSD) [81], which originally had $\sim 243,000$ materials. We cleaned this dataset by: 1) removing non-symmetry structures, 2) filtering out semiconductors using band gap data from the Materials Project [82], and 3) eliminating any materials already in SuperCon. The final set contained 89,491 well-defined, synthesized crystal structures. Crucially, this cleaned ICSD data was completely unseen during training, allowing us to fairly test the model’s ability to discover new superconductors across all material types. Also the train and test sets were split in a way that preserves the percentage of samples for each class (80/20).

2.1. Classification framework

Superconductors in the training dataset are categorized into conventional and unconventional types based on their pairing mechanisms. Conventional superconductors follow the BCS theory [83], where phonon-mediated interactions enable cooper pair formation, typically resulting in $T_c < 25$ K. Exceptions like MgB_2 ($T_c \approx 39$ K) [84] and high-pressure hydrides (e.g., H_3S , $T_c > 200$ K) [85,86] are noted. Unconventional superconductors, such as cuprates and iron-based compounds, exhibit higher T_c values and rely on non-phonon mechanisms

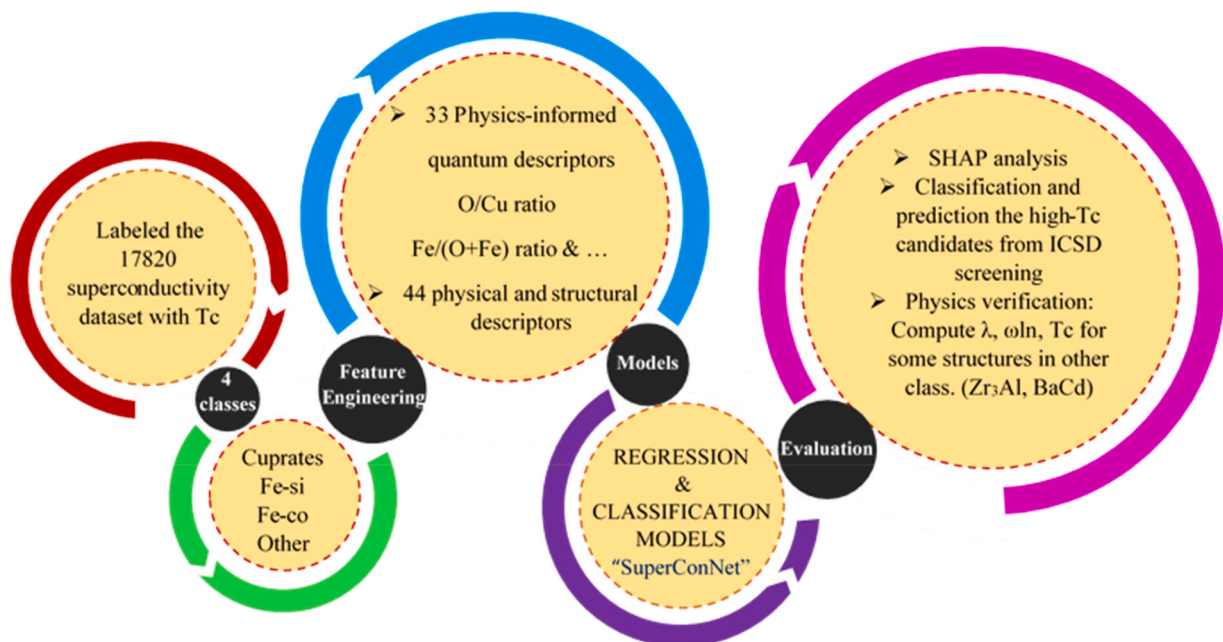


Fig. 1. Workflow of predicting classes of superconductivity and the transition temperatures of synthesized superconducting materials.

like spin fluctuations. To refine predictions, we divided the training dataset into four subclasses: a. CuO: cuprates (e.g., $\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_{8+\delta}$, $T_c \approx 133$ K) [87], defined by CuO_2 layers critical for superconductivity; b. Fe-co: complex iron-based superconductors (e.g., $\text{LaFeAsO}_{1-x}\text{F}_x$, $T_c \approx 56$ K) [88] with multi-element structures; c. Fe-si: simple iron-based superconductors (e.g., FeSe , bulk $T_c \approx 8$ K) [89]; and d. Other: other superconductors, including conventional and less-studied unconventional types [90]. This hierarchy ensures that structural and electronic distinctions are captured, thereby enhancing model interpretability (see Supporting Information).

2.2. Feature engineering and model training

Several known structural features influence the transition temperature of cuprate and iron-based superconductors. Examples include the number of CuO_2 planes, the buckling of CuO_2 planes, the $\text{Cu-O}_{\text{apex}}$ distance, the vertical distance (z) of the pnictogen (P/As/Sb) or chalcogen (S/Se/Te) atom above or below the Fe plane, the Fe–As–Fe bond angle, the Fe–Fe distance, the distance between pnictogen atoms across the Fe plane, the distance between FeAs layers, and the ratio of in-plane (a) to out-of-plane (c) lattice parameters, doping level, magnetic impurities, among others (Supporting Information) [91–94]. These features are common among all superconductors and provide knowledge of the exact crystal structure of the compound. Instead, for each compound, elemental properties (e.g., electronegativity, atomic radius, etc.; see Supporting Information) were weighted by composition to generate feature vectors. After feature selection, 44 key descriptors linked to superconductivity and 33 PI features were retained (Supporting Information).

3. Results and discussion

3.1. Machine learning workflow

Our approach to predicting synthesized superconducting materials involves a strategic two-stage process: first, classifying ICSD materials into distinct categories and then predicting their critical transition temperatures. We analyzed approximately 17,820 experimental superconducting records, organized into four classes: 8540 cuprates (copper-oxide-based materials), 296 simple iron-based (Fe-si) compounds, 113 complex iron-based (Fe-co) systems, and 8871 other unconventional

superconductors (Fig. 2a). From an initial 33 PI and 44 compositional descriptors (Supporting Information), we selected 59 descriptors by feature engineering that demonstrated maximum discriminative power while maintaining computational robustness, as the excluded features exhibited problematic characteristics, including uniform values across materials, excessive zero entries, or undefined parameters in significant portions of the dataset. The critical descriptors include the cuprate signature, identified through the CuO_2 plane count, which serves as a fingerprint for copper-oxide superconductors like YBCO and BSCCO by quantifying oxygen coordination in copper-oxygen planes. Optimal T_c occurs at 3–4 planes where quantum confinement maximizes Cooper pair coherence. The iron-based marker utilizes absolute iron content to differentiate iron-based superconductors from other classes, where simple systems (Fe-si) such as FeSe show high iron density with minimal oxygen, while complex systems (Fe-co) like LaFeAsO incorporate iron-pnictogen interactions, with zero-iron materials automatically excluded from both categories. The iron subtype separator utilizes the $\text{Fe}/(\text{O} + \text{Fe})$ ratio to distinguish between iron-dominated structures (Fe-si class, ratios >0.7 , where superconductivity emerges directly from iron layers) and complex systems (Fe-co class, ratios 0.2 – 0.5 , involving iron-oxygen/pnictogen interactions). The unconventionality indicator leverages magnetic properties to identify materials where antiferromagnetic fluctuations mediate electron pairing in cuprates and iron-based systems, contrasting with conventional superconductors showing no magnetic ordering (Supporting Information). The conventionality detector utilizes electron-phonon coupling (λ) to distinguish between phonon-mediated superconductors in the “other” class ($\lambda > 1$, as in MgB_2 or Nb_3Sn) and exotic mechanisms in cuprates and iron-based materials ($\lambda < 0.3$) (Fig. 3a). Finally, the complexity gauge tracks transition metal density, capturing how additional elements, such as zinc, suppress superconductivity in cuprates [95], while cobalt doping enhances critical temperatures in iron-based systems [96]. This feature quantifies chemical complexity and its impact across all classes (Supporting Information).

The PI descriptors work synergistically with 44 structural and physical features (Supporting Information) to achieve high classification accuracy, as they encode some aspects of the quantum mechanical conditions required for superconductivity. The confusion matrix demonstrates their power, with a clear separation between classes and minimal overlap, particularly in distinguishing cuprates from iron-based materials and conventional from unconventional systems. The selected

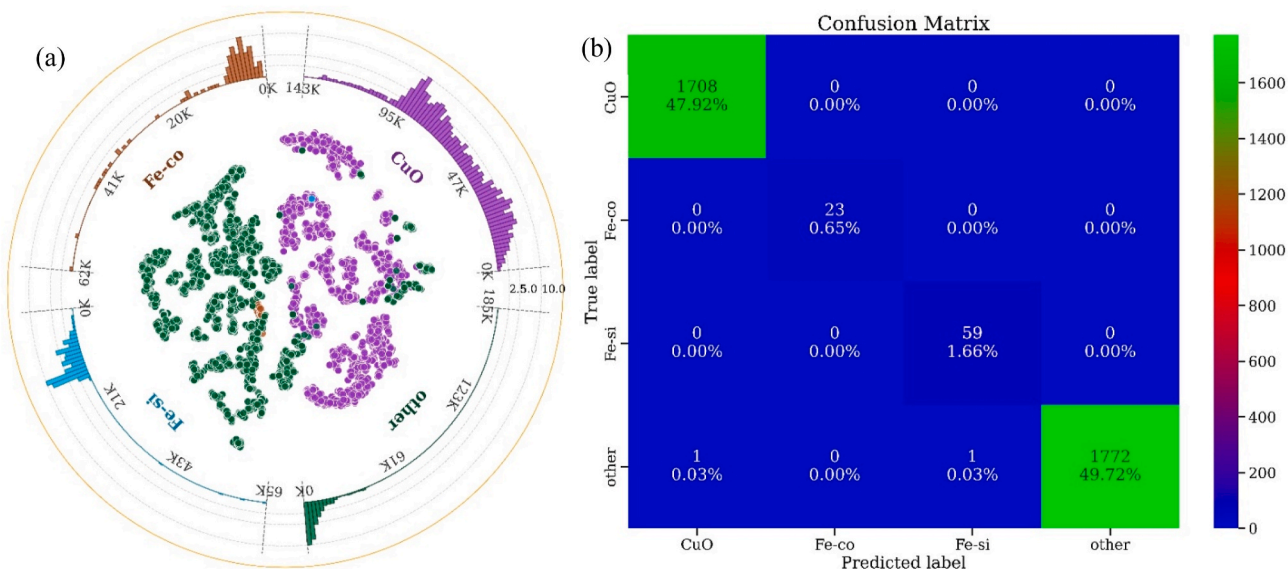


Fig. 2. a) (outside) The distribution of the training dataset that is divided into four classes. (inside) The t-SNE projection of all four classes in two dimensions. b) The confusion matrix of the model on test structures.

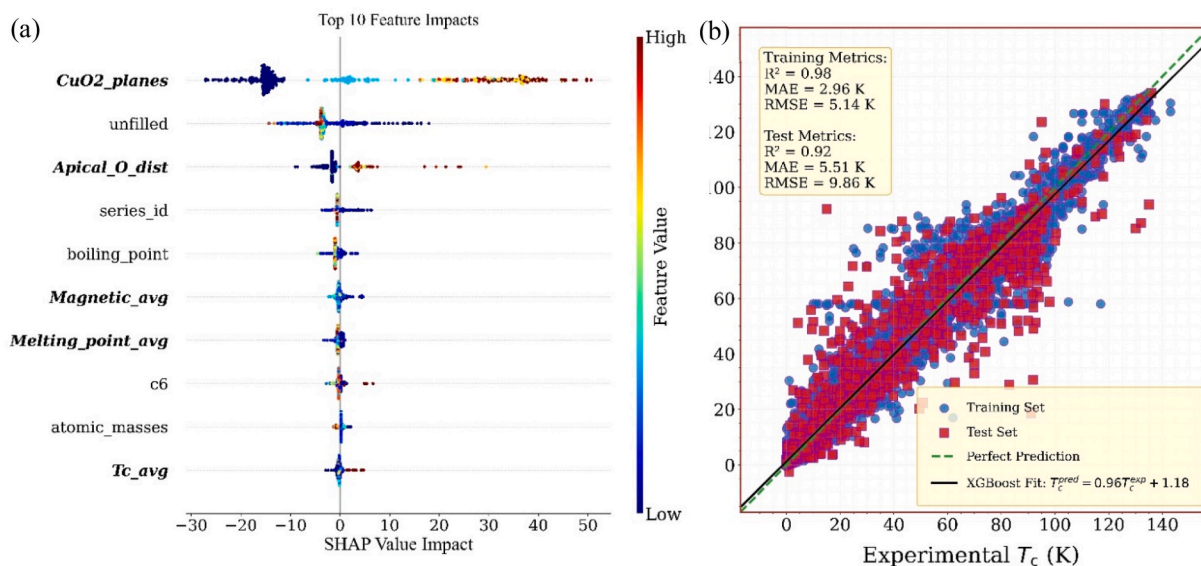


Fig. 3. a) The SHAP values of the 10 top features on the output results for all classes. b) The prediction results for the transition temperature of superconductivity based on the XGBoost regression model.

59 features not only predict but also explain why certain materials become superconducting while others don't, bridging materials chemistry with quantum physics in a way that purely structural parameters cannot. By focusing on these fundamental indicators, our model captures the essential physics governing all known high-temperature superconductors. The classification model demonstrated exceptional performance in categorizing superconductors into four distinct classes: cuprates (CuO), simple iron-based (Fe-si), complex iron-based (Fe-co), and other (un)conventional materials, achieving near-perfect accuracy (99.94 %) and F1-score (99.94 %) across 3564 samples. The confusion matrix revealed consideration separation between classes, with zero misclassifications for the 1708 cuprate samples and only two errors among 1774 other superconductors (Fig. 2b). Notably, the model's ability to accurately identify 23 complex iron-based compounds, despite their limited representation, underscores its capacity to capture subtle distinguishing features. This robust classification performance validates our class definitions as physically meaningful categories. It confirms fundamental differences in superconducting mechanisms between material families, especially the clear separation between copper-oxide and iron-based systems.

Transitioning to critical temperature prediction, our regression approach employed XGBRegressor, random forest, and TabNet models (Supporting Information) trained on approximately 17,820 experimental records and tested on unseen ICSD data. By removing the SuperCon component from the ICSD dataset, the XGBRegressor achieved best accuracy with training metrics ($R^2 = 0.98$, MAE = 2.96 K, RMSE = 5.14 K) and maintained strong generalization on the test data ($R^2 = 0.92$, MAE = 5.51 K, RMSE = 9.86 K). This narrow performance gap indicates minimal overfitting, with prediction errors within ± 4 K for typical cuprates, a clinically relevant precision for materials screening. The TabNet model exhibited slightly lower but still robust performance (training $R^2 = 0.89$, MAE = 6.92 K; validation $R^2 = 0.89$, MAE = 6.94 K), reflecting the inherent trade-off between interpretability and precision, while confirming the universal predictive power of our feature set. Both models consistently outperformed conventional approaches, demonstrating applicability across diverse superconductor families (Fig. 3b–S2, S3, S4).

To understand the contributions of individual features to the regression model's predictions, we conducted a SHAP (SHapley Additive exPlanations) [97] analysis. The SHAP analysis offers valuable insights into the model's decision-making process, enabling us to better

understand which features are most crucial for achieving accurate regression results. For all classes (Fig. 3a), the SHAP values identified CuO₂_planes as the most influential feature with a mean absolute SHAP value of 19.93, signifying its dominant role in predictions. This was followed by unfilled (4.00) and Apical_O_dist (2.61), with five physics-informed features appearing in the top 10: CuO₂_planes (1st), Apical_O_dist (3rd), Magnetic_avg (6th), Melting_point_avg (7th), and Tc_avg (10th). The prominence of CuO₂_planes and Apical_O_dist highlights the critical role of cuprate-specific structural configurations, where the geometry and count of superconducting planes directly modulate electronic properties. Magnetic_avg emerged as a key mediator of unconventional pairing mechanisms, particularly in iron-based systems where antiferromagnetic fluctuations govern superconductivity. The appearance of Melting_point_avg and Tc_avg suggests the model leverages thermal stability trends and historical critical temperature data to contextualize predictions. These physics-informed features collectively encode fundamental quantum mechanical principles: CuO₂_planes and Apical_O_dist quantify structural determinants of hole doping in cuprates, while Magnetic_avg captures spin-fluctuation dynamics essential for electron pairing in iron-based superconductors. Their dominance over non-physics features (e.g., unfilled and boiling_point) underscores the model's reliance on domain-specific mechanisms. Notably, Tc_avg's inclusion indicates the model incorporates prior knowledge of critical temperature trends across material families, enhancing predictive coherence. The synergy between these features, particularly the coupling of structural descriptors (CuO₂_planes) with electronic/magnetic properties (Magnetic_avg), reveals nonlinear interactions that govern phase competition between superconductivity and magnetic ordering. This integration of physics-based features surpasses 44 compositional descriptors in global impact, providing a robust framework for identifying high- T_c candidates. The exceptional performance of both classification and regression models stems from these features' ability to isolate material-specific pairing mechanisms, structural dimensionality for cuprates, and magnetic mediation for iron-based systems while quantifying universal trends in thermal and electronic behavior. In this way, the model translates complex quantum phenomena into a set of interpretable predictors, contributing to targeted feature engineering for the design of superconducting materials.

3.2. First principles calculation

To validate our machine learning model, we performed first-principles electron-phonon coupling calculations on some small and binary BCS-type superconductors. Our DFT calculation results indicate that a few of these materials are dynamically unstable. Here, we only present the results for two of them (BaCd [98] and Zr₃Al [99–101]), which were previously synthesized, and the band structure, phonon dispersions reveal the metallic and dynamic stability of these two structures (Fig. 4a,b,c,d), in which the transition temperatures were not considered. The BaCd compound may exhibit low-temperature superconductivity ($T_c < 3$ K), similar to mercury-alkaline earth metal systems [102] or mercurides of strontium compounds [103]. On the other hand, the Zr₃Al structure is similar to the well-known A15 superconductors, many of which exhibit superconductivity around 20 K [104]. Due to the challenges in calculating the electron-phonon coupling constant in metals, multiple double-delta smearing values (0.005–0.05 Ry) were employed in the Quantum ESPRESSO code to ensure convergence. For BaCd (Fig. 5a), the Eliashberg spectral function $\alpha^2F(\omega)$ extends from 4 to 10 meV without a separation gap between peaks. This indicates that all phonon modes contribute to the electron-phonon interaction. This behavior leads to an electron-phonon coupling constant $\lambda \approx 0.5$ for a smearing value of 0.05 Ry, with a corresponding logarithmic average phonon frequency ($\omega_{\log}$) of 75.5 K. The computed critical temperature is approximately 0.9 K. The inset indicates that T_c converges for reasonable values of the smearing parameter. Thus, the T_c value for this compound falls within the expected range ($T_c < 3$ K) observed in mercury-alkaline earth metal systems. Fig. 5b shows the Eliashberg spectral function $\alpha^2F(\omega)$ for the Zr₃Al compound. The maximum phonon frequency here is nearly three times larger than in BaCd, consistent with

the lighter atomic masses of Zr/Al versus Ba/Cd. The $\alpha^2F(\omega)$ spectrum divides into two regions: The primary contribution to λ originates from phonon modes below ~ 26 meV, while higher-frequency modes ($\omega > 29$ meV) contribute minimally, yielding a cumulative $\lambda \approx 0.56$. For a smearing value of 0.05 Ry, $\omega_{\log} \approx 190.6$ K approximately 2.5 times greater than in BaCd. These parameters yield $T_c \approx 3.5$ K. However, this critical temperature remains notably lower than those of high- T_c A15 superconductors (e.g., Nb₃Sn, $T_c \approx 18$ K). Given the similar λ values in Zr₃Al and BaCd, the higher T_c in Zr₃Al relative to BaCd (0.9 K) likely arises from its enhanced phonon frequencies, though both systems exhibit substantially lower T_c than optimized A15 compounds.

The machine learning model predicted T_c values of 4.47 K for BaCd and 5.47 K for Zr₃Al. Therefore, our model predicts that the transition temperature of these compounds is approximately 3 K higher than the prediction of the BCS model. However, applying the XGBoost established regression relationship ($T_{c(\text{pred})} = 0.96T_{c(\text{exp})} + 1.18$) reconciles this apparent discrepancy. For BaCd, the regression-adjusted experimental equivalent is 3.42 K $[(4.47 - 1.18)/0.96]$, which falls around the computationally predicted range (4.47 ± 5.51 K as the MAE) and approaches the range of computational bound. Similarly, for Zr₃Al, the adjusted value of 4.47 K $[(5.47 - 1.18)/0.96]$ is approximately upper the BCS model (Fig. 5b). This divergence suggests either limitations in the electron-phonon calculations for Zr₃Al or the presence of additional superconducting mechanisms beyond conventional BCS theory in this material. However, both methods predict that the BaCd and Zr₃Al compounds are superconducting candidates.

The observed discrepancies provide valuable insights into the strengths and limitations of both approaches. The underprediction by electron-phonon methods likely stems from known limitations in density functional theory for capturing anisotropic phonon modes or

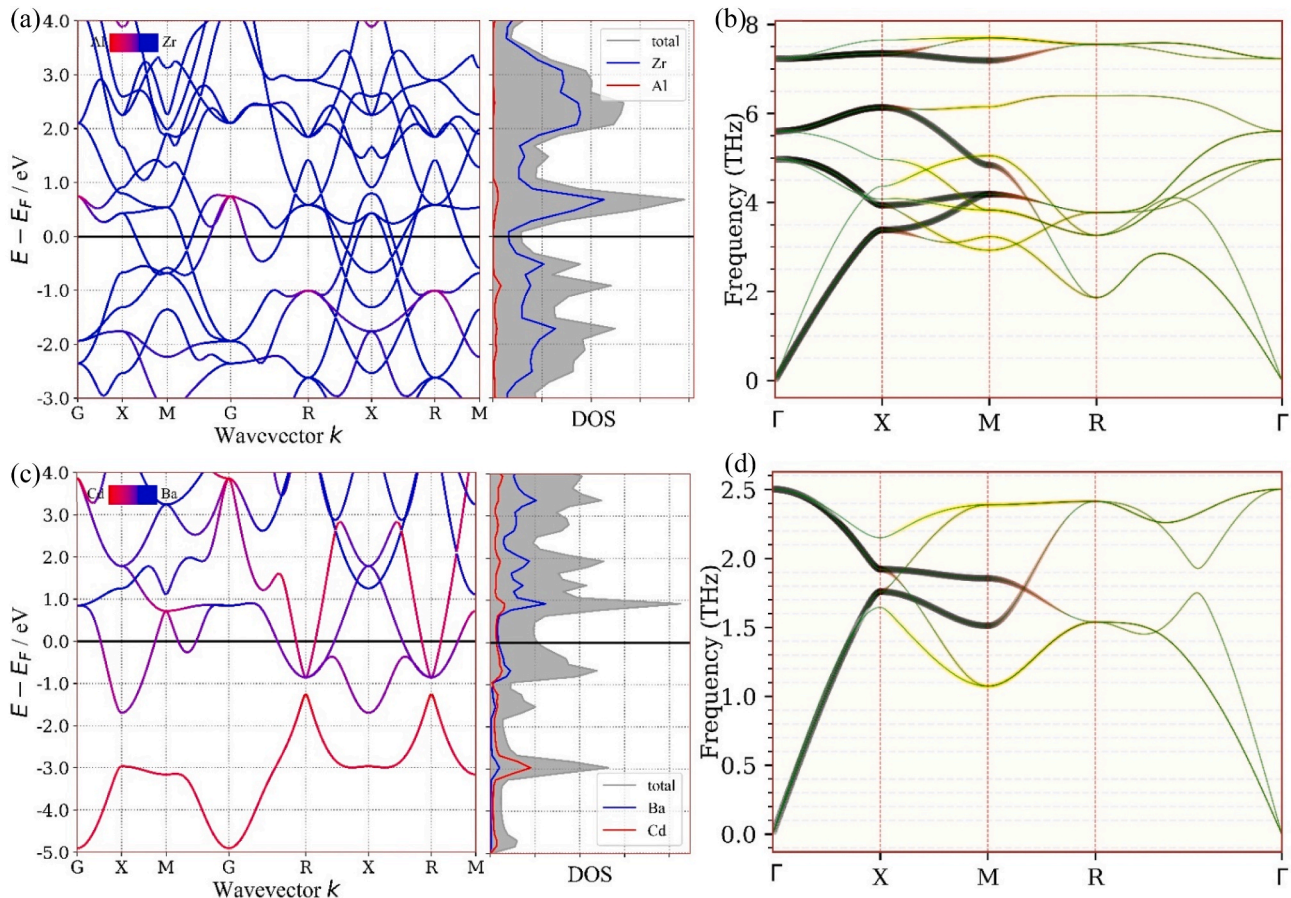


Fig. 4. The band structure, the density of states, and phonon dispersions for the (a, b) Zr₃Al and (c, d) BaCd. The Fermi is at zero.

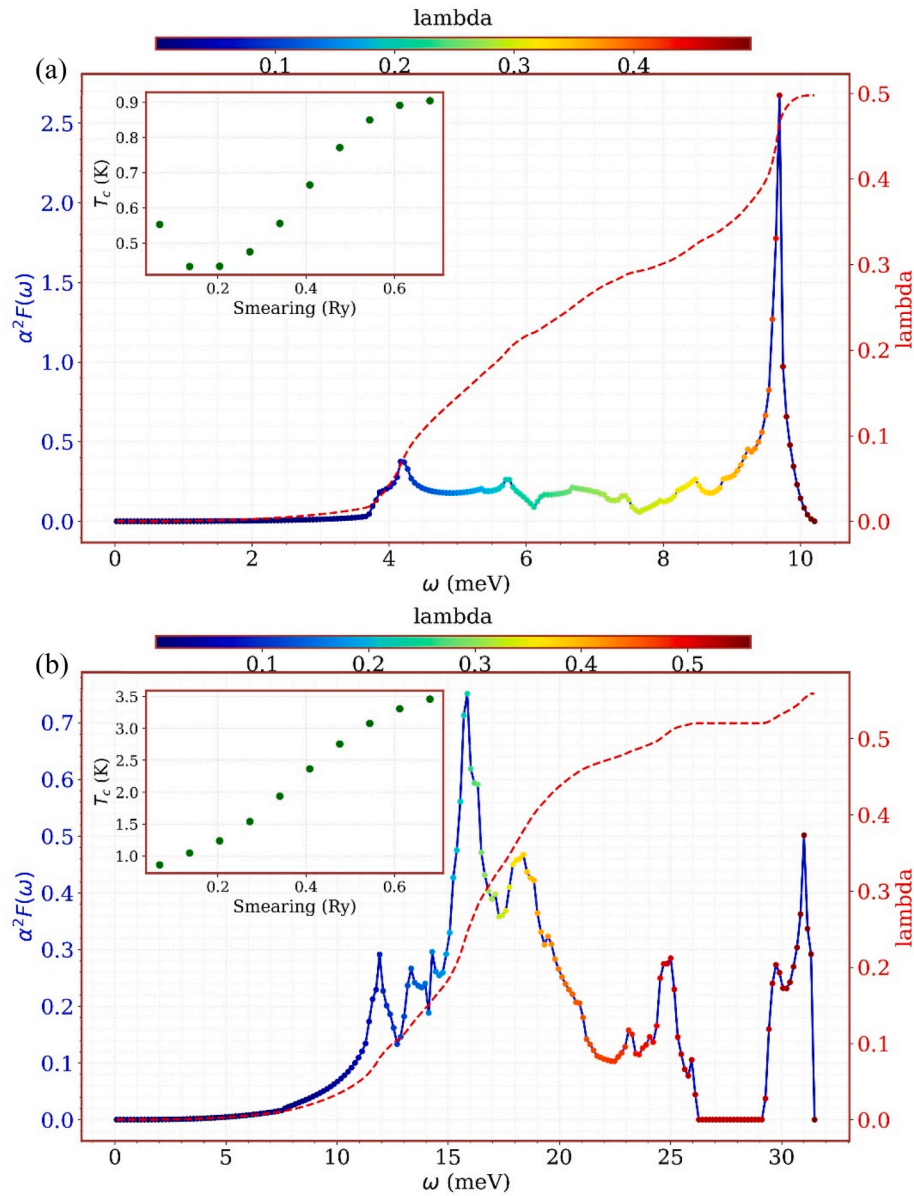


Fig. 5. Relation between EPC function vs. frequency and transition temperature (T_c) vs electron-phonon coupling parameters of a) BaCd and b) Zr₃Al, the two potential candidate superconductors.

anharmonic effects. Furthermore, features such as multi-band Fermi surfaces, which can enhance superconductivity beyond isotropic phonon-mediated mechanisms, may be present in this material. Our machine learning model, trained on a diverse dataset of superconducting materials, appears capable of capturing these contributing factors. Our machine learning (ML) framework was designed to incorporate indicators for both conventional and unconventional superconductivity. The model accounts for physical phenomena that can be challenging to capture with standard computational methods, especially in materials with complex electronic structures. The ability to generate reasonable T_c estimates for novel materials without expensive first-principles calculations represents a significant practical advantage. The difference in predicted transition temperature may arise from physical parameters not routinely captured by first-principles calculations but implicitly incorporated within the ML model. This cross-validation methodology establishes a powerful paradigm for superconductivity research, combining the interpretability of physical models with the predictive power of machine learning. The approach enables the rapid screening of candidate materials while providing insights into

the underlying quantum mechanisms that govern their superconducting behavior.

In cuprates and iron-based superconducting structures, the approach accurately forecasts critical temperatures by analyzing fundamental physical descriptors. The model identifies some structures, such as Sr₄Mn₂Cu₂S₂O₆, as a promising cuprate (predicted $T_c = 54.8$ K) through optimal PI descriptors: two CuO₂ planes enabling 2D superconductivity, apical oxygen distance (2.35 Å) facilitating efficient charge transfer, and hole doping (0.86) near the quantum-critical regime, validating the same feature hierarchy that governs record-setting compounds like Ba₂Ca₂Cu₃Hg₁O₈. For iron-based systems, the precise tuning of Fe-layer separation (~ 5.2 Å) and magnetic coupling (0.16–0.17) in predicted superconducting structures confirms spin-fluctuation pairing can be engineered via PI-guided design. Crucially, these predictions extend beyond known materials: our screening of the ICSD database (Supporting Information) reveals some cuprates and iron-based candidates with $T_c > 50$ K, all sharing three notable PI characteristics (1) dimensional coherence (CuO₂ planes ≥ 3 or Fe-layer separation ≤ 5.3 Å), (2) magnetic/electronic synergy (weighted magnetic moment $> 1.8\mu_B$ or

electron concentration $>6.5e^-/\text{atom}$), and (3) thermodynamic stability (melting point >500 K). Our approach moves beyond statistical correlations by embedding causal physical mechanisms directly into the model architecture. This proves that high- T_c superconductivity stems from controllable atomic-scale conditions, not accidental chemical combinations. By quantifying how specific structural parameters (e.g., CuO_2 layer count, Fe-Fe spacing) determine superconducting performance, we transform materials discovery from trial-and-error to physics-guided engineering.

3.3. Summary

This study establishes a machine learning framework for superconductor discovery, integrating stoichiometry descriptors besides 33 physics-informed descriptors, CuO_2 layer dimensionality (cuprates), Fe-Fe spacing (iron-based systems), electron-phonon coupling (λ) (conventional superconductors), magnetic properties, and transition-metal complexity to achieve near-perfect classification (99.94 % accuracy) via SuperConNet, our custom residual network engineered with physics-aware normalization and mechanism-specific skip connections for unprecedented class separation. For regression, TabNet demonstrated superior interpretability using attention-based feature selection and sparsity regularization ($\lambda_{\text{sparse}} = 1e^{-4}$), achieving robust critical temperature prediction (MAE = 6.94 K, $R^2 = 0.89$) despite slightly lower precision than XGBoost, excelling in capturing complex quantum interactions via its stepwise explainable architecture. The model unifies prediction across all superconductor classes, eliminating family-specific approaches, while SHAP analysis decodes quantum mechanisms linking descriptors like magnetic properties to transition temperatures. Validation against DFT/BCS calculations exposed limitations in traditional methods (e.g., T_c deviations in Zr_3Al), highlighting physics-informed ML's ability to capture elusive parameters. Screening ICSD materials proves scalable real-world discovery, reducing computational costs and enabling resource-efficient candidate identification. By bridging interpretable ML with quantum phenomena, this framework pioneers a paradigm shift in materials exploration, extendable to correlated-electron systems like topological superconductors, and accelerates ML-guided synthesis of next-generation high- T_c materials. This work establishes a unified predictive framework for ambient-pressure superconductors, thereby mapping a clear path for future expansion to high-pressure phases as standardized data becomes available.

4. Methods

4.1. Machine learning training

For regression tasks predicting superconducting transition temperatures (T_c), three methods were employed: Random Forest utilized bootstrap aggregation of decision trees to robustly model feature interactions; XGBoost implemented gradient boosting with sequential error correction for enhanced predictive accuracy; and TabNet leveraged attention-based neural networks for interpretable feature selection. For classification of superconductor families, we developed SuperConNet, a novel Custom Residual Network architecture specifically engineered for material categorization. This approach features specialized residual connections and quantum descriptor-informed regularization to achieve near-perfect family separation. Complete architectural details, hyperparameter specifications, and computational frameworks for all methods are provided in the Supporting Information. The SuperConNet design incorporates physics-aware normalization layers and mechanism-specific skip connections to distinguish subtle quantum mechanical differences between superconductor classes, representing a significant advancement in materials informatics methodology.

4.2. Density functional theory (DFT) calculations

This study employs density functional theory (DFT) calculations performed within the Quantum ESPRESSO (QE) and VASP code (Supporting Information). Here, we utilized a plane-wave basis set combined with SG15 Optimized Norm-Conserving Vanderbilt (ONCV) pseudopotentials. Exchange-correlation effects were modeled using the Generalized Gradient Approximation (GGA) with the Perdew-Burke-Ernzerhof (PBE) functional. Plane-wave kinetic energy and charge density cutoffs were set to ≥ 100 Ry and ≥ 400 Ry, respectively. Convergence was rigorously enforced, with thresholds set below 10^{-5} Ry for total energy (ionic minimization), 10^{-12} Ry for self-consistency, and 10^{-4} Ry/au (< 3 meV/Å) for forces (ionic minimization).

Density Functional Perturbation Theory (DFPT) was employed to calculate the dynamical matrix and electron-phonon coupling (EPC) parameters via the linear response approach. Charge density distributions were computed using a $12 \times 12 \times 12$ k-point mesh. For phonon properties and EPC evaluation, a minimum $4 \times 4 \times 4$ q-point grid was adopted, while electron-phonon interaction matrices were resolved with a denser $48 \times 48 \times 48$ k-point grid. Electron-phonon coupling properties and superconducting critical temperature (T_c) were analyzed within the Bardeen-Cooper-Schrieffer (BCS) framework using McMillan's equation. The Eliashberg spectral function, denoted $\alpha^2F(\omega)$, quantifies the frequency-dependent electron-phonon interaction strength and relates directly to the EPC constant λ . This function is defined as follows:

$$\alpha^2F(\omega) = \frac{1}{2\pi N(E_F)} \sum_{qv} \delta(\omega - \omega_{qv}) \frac{\gamma_{qv}}{\hbar\omega}$$

where $N(E_F)$ is the electronic density of states (DOS) at the Fermi level, ω_{qv} represents the frequency of phonon mode v for wave vector q , and γ_{qv} denotes the phonon linewidth – a measure of electron-phonon coupling strength for mode qv . This linewidth is determined by:

$$\gamma_{qv} = 2\pi\omega_{qv} \sum_{ij} \frac{d^3k}{\Omega_{\text{BZ}}} |g_{qv}(k, i, j)|^2 \delta(E_{k,i} - E_F) \delta(E_{k+q,j} - E_F)$$

where Ω_{BZ} is the Brillouin zone (BZ) volume, $E_{k,i}$ and $E_{k+q,j}$ are Kohn-Sham energies at states (k, i) and $(k + q, j)$, and $g_{qv}(k, i, j)$ is the screened electron-phonon matrix element. The mode-resolved electron-phonon coupling constant λ_{qv} is then given by:

$$\lambda_{qv} = \frac{\gamma_{qv}}{\pi\hbar N(E_F)\omega_{qv}^2}$$

The total electron-phonon coupling constant λ is determined by Brillouin-zone averaging of the mode-resolved couplings λ_{qv} :

$$\lambda = \sum_{qv} \lambda_{qv} = 2 \int_0^\infty \frac{\alpha^2F(\omega)}{\omega} d\omega$$

Thus, the total coupling constant λ equals twice the first inverse moment of the Eliashberg spectral function $\alpha^2F(\omega)$. The superconducting critical temperature T_c is subsequently derived from the modified McMillan equation:

$$T_c = \frac{\omega_{\text{ln}}}{1.2} \exp\left(\frac{-1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)}\right)$$

Within this equation, the Coulomb pseudopotential μ^* is fixed at 0.1. The characteristic logarithmic phonon frequency ω_{ln} is subsequently defined as:

$$\omega_{\text{ln}} = \exp\left(\frac{2}{\lambda} \int_0^\infty \ln(\omega) \frac{\alpha^2F(\omega)}{\omega} d\omega\right)$$

CRediT authorship contribution statement

Ehsan Alibagheri: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Sandoghchi:** Writing – review & editing, Investigation, Formal analysis. **Alireza Seyfi:** Writing – original draft, Data curation. **Mohammad Khazaei:** Writing – review & editing, Software. **S. Mehdi Vaez Allaei:** Supervision, Software, Project administration, Writing – review & editing.

Code availability

The source code used in the current study is available at GitHub https://github.com/ehsanab550/PIML_superconductivity.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mtphys.2025.101971>. Contains the SuperCon training dataset, which is labeled into four classes, and the predicted ICSD results

Data availability

The data supporting this study's findings are available in this article's supplementary material. Contains the SuperCon training dataset, which is labeled into four classes, and the predicted results.

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